TECHNICAL EVALUATION REPORT: RANDOM FOREST MODEL

Detailed Specification & Multi-Circuit Performance Benchmark for Edoardo's Pipeline (Sprint 3 Task 6)

Model Owner: ML_RandomForest | **Prepared By:** Data Science Team (Task 6 Lead) | **Date:** November 2026

1. MODEL ARCHITECTURE & HYPERPARAMETERS

The model utilizes a pooled ensemble architecture based on `sklearn.ensemble.RandomForestRegressor`. It is trained without circuit identity, making it purely feature-driven and capable of zero-shot transfer learning across unseen energy circuits.

- **Scaling & Preprocessing:** Each feature value is divided by its circuit's training mean prior to fitting; predictions are re-scaled back post-inference.
- **+1h Horizon:** `criterion='absolute_error'`, `n_estimators=300`, `max_depth=12`, `min_samples_leaf=100`, `max_features=1.0`.
- **+6h & Day-Ahead Horizons:** `criterion='absolute_error'`, `n_estimators=300`, `max_depth=4`, `min_samples_leaf=250`, `max_features=0.5`.
- **Global Settings:** `bootstrap=True`, `random_state=0`.

2. FEATURE ENGINEERING & REPRODUCIBILITY

Uses 15 standardized features from `task8_model.FEATURES` including lagged consumption values, rolling statistics, and calendar embeddings (hour, day of week, weekend indicator).

```
Repository: https://github.com/TBDA-
Classes/2026G2/commit/0d882d79f9b39a87b726e0f75faf7799d3f8413a
Package: Model3_RandomForest.zip (main branch, commit 0d882d7)
Script: step4_test.py -> Output: RandomForest_test_predictions.csv (8,064 rows)
Environment: Python 3.13, scikit-learn 1.9.1, pandas 3.0.5, numpy 2.5.3
```

3. PERFORMANCE RESULTS (16 CIRCUITS)

Evaluated on 8,064 total prediction rows across 16 circuits (8 trained NE/NW + 8 unseen SE/SW) over the fixed held-out test window (29 Sep – 6 Oct 2025 UTC):

Horizon	Evaluation Scope	MAE (Wh)	RMSE (Wh)	Performance & Zero-Shot Note
+1h	Trained Circuits (NE/NW)	35.9	141.8	Competitive short-term tracking
+1h	Unseen Circuits (SE/SW)	30.1	118.4	Solid zero-shot performance
+6h	Trained Circuits (NE/NW)	47.1	149.3	Top-performing medium-term ML model
+6h	Unseen Circuits (SE/SW)	31.7	122.5	Significantly outperforms linear models
Day-Ahead	Trained Circuits (NE/NW)	46.7	148.1	Lowest day-ahead MAE overall
Day-Ahead	Unseen Circuits (SE/SW)	32.1	124.2	Significantly outperforms linear models

4. VISUAL EVALUATION & ERROR BOUNDS



Figure 1: MAE evaluation with 2,000-resample 24h block bootstrap 95% confidence intervals across horizons.

5. MODEL LIMITATIONS & RECOMMENDATIONS

- **Strengths:** Strong generalization across unseen circuits; non-linear peak capturing at medium and long horizons (+6h and Day-Ahead).
- **Limitations:** Tree-based regressors cannot extrapolate demand spikes exceeding training maximums during severe anomalous events. Evaluated on one test week; SE/SW circuits have low sensor coverage (34.5% – 88.7%).
- **Recommendation:** Adopt Random Forest as the core engine for +6h and Day-Ahead forecasting across all energy circuits.